

Three Modes for Simulated Risk: A Data-Driven Spectral Basis for Portfolio Loss Distributions

Tamás Nagy, Ph.D.

ORCID: 0009-0004-8079-4679

Independent Researcher

tnagyphd@gmail.com

<https://thalens.org>

Working Paper • August 2026

Build-std v2.0+d27f3744d6 | 2026-08-10 12:46 | 18d419bf

Abstract

The Spectral Fenton Distribution represents a portfolio loss distribution with 128 Fourier coefficients. Under the deterministic simulated protocol reported here (5 assets, 250 trading days, seed 42), a single principal component captures 97.81% of observed coordinate variation and three components capture 99.9996%. Three coordinates — $z_1(t)$, $z_2(t)$, $z_3(t)$ — provide a $43\times$ per-day coefficient compression with a maximum relative VaR reconstruction error of 0.0416%. A basis fitted on days 1–200 has a maximum relative VaR error of 0.0189% on days 201–250. The dominant mode z_1 is strongly associated with simulated VaR; economic interpretations of the remaining modes are tentative. The coordinates provide a descriptive reduced-order representation, not a calibrated detector or an empirical claim about real portfolios.

Key Messages

- 128 spectral coefficients compress to 3 numbers with $<0.05\%$ VaR error (simulated data, $n = 5$ assets)
 - Mode 1 alone captures 97.8% of observed simulated coefficient variation; three modes capture 99.9996%
 - A 200-day fitted basis gives at most 0.0189% relative VaR error on the 50-day simulated holdout
 - The 3-coordinate dashboard visualizes simulated stress episodes via Mahalanobis distance; detection performance remains untested
 - The learned basis offers a candidate reduced-order representation for Dynamic URRT, Bayesian Risk, and optimal transport
-

1. Introduction

1.1 The Dimensionality Question

The Spectral Fenton Distribution (Nagy, 2026, `fin_fenton_solved`) encodes a portfolio’s loss distribution as $N = 128$ Fourier-cosine coefficients $A_0, \dots, A_{127}$. This is already a dramatic compression from the infinite-dimensional space of probability distributions. But for real-time risk management, even 128 numbers may be too many: a risk manager needs to understand, at a glance, what changed today.

The question is: **how many independent ways can the distribution actually change?**

If the answer is 3, then the risk dashboard is three numbers. If the answer is 10, it is ten. The answer determines the intrinsic dimensionality of the risk monitoring problem.

1.2 The PCA Approach

Let $A(t) \in \mathbb{R}^{128}$ denote the coefficient vector on day t . Over T trading days, we observe a $T \times 128$ matrix. Principal Component Analysis (PCA) extracts the dominant modes of variation:

$$A(t) \approx \bar{A} + z_1(t) \cdot v_1 + z_2(t) \cdot v_2 + \dots + z_r(t) \cdot v_r$$

where $\bar{A}$ is the time-averaged coefficient vector, $v_1, \dots, v_r$ are orthonormal directions in $\mathbb{R}^{128}$ (the principal modes), and $z_1(t), \dots, z_r(t)$ are the coordinates on day t . The question reduces to: how large must r be?

1.3 Related Work

PCA is a foundational tool in quantitative finance. The seminal application to interest-rate term structures by Litterman and Scheinkman (1991) showed that three factors — level, slope, and curvature — capture most yield-curve variation. Dai and Singleton (2000) connected low-dimensional factor structure to affine term-structure models. The present study changes the object being decomposed: each observation is a coefficient vector encoding a simulated portfolio loss distribution rather than a vector of yields or asset returns.

Our work differs in *what* is being decomposed: not asset returns, not yield curves, but the Fourier-cosine coefficients that encode a portfolio loss distribution. This is a finite-dimensional SVD on the coefficient matrix. Because the integration domain may vary with the daily cumulant rule, the analysis is an empirical comparison of coefficient coordinates rather than a fixed-function-space PCA; the daily domain is additional state. The key empirical finding is that 3 modes provide near-perfect reconstruction of the reported simulated coefficient trajectories, which parallels the yield curve result but concerns distributional coordinates rather than price dynamics.

1.4 Contribution

We find empirically that $r = 3$ captures 99.9996% of observed simulated coefficient variation across 250 trading days:

1. **Compression:** $128 \rightarrow 3$ numbers (43 $\times$) with mean VaR error 0.005% (Section 3).
2. **Interpretation:** Mode 1 is strongly associated with VaR ($r = 0.998$); the interpretations of Modes 2–3 are tentative (Section 3.3).

3. **Dashboard:** each day is a point $(z_1, z_2, z_3) \in \mathbb{R}^3$. Mahalanobis distance visualizes simulated outliers, while detection performance remains open (Section 3.4).
4. **Holdout diagnostic:** the three-mode representation remains accurate on the declared chronological simulation holdout (Section 3.6); cross-portfolio and crisis robustness are outside this version’s evidence.
5. **Unification:** the learned basis offers candidate reduced-order coordinates for Bayesian filtering (#7), temporal compression (#2), Schrödinger bridges (#11), and instanton paths (#13), subject to model-specific approximation validation.

This study is a companion to *Deterministic Log-CF Evaluation for Correlated Lognormal Sums, Exact Portfolio VaR Without Monte Carlo: The Eigen-COS Method, Noise-Free Risk: Deterministic VaR, ES, and Spectral Risk Measures*, and *The Anomaly Functional: Real-Time Arbitrage Detection via Spectral Risk Coefficients* (Nagy, 2026). These related working and Zenodo papers provide the distribution, deterministic-risk, and anomaly contexts; they do not certify the numerical results reported here.

2. Method

2.1 Data Generation

We simulate $T = 250$ trading days of an $n = 5$ -asset portfolio with random seed 42. The implemented update is a discrete stochastic recurrence, not a calibrated continuous-time SDE. Initial volatilities are deterministic, $\sigma_i(0) = 0.15 + 0.1i/5$ for $i = 0, \dots, 4$, and the equal portfolio weights remain fixed.

At each day, the code applies

$$\sigma_i \leftarrow \text{clip}(\sigma_i \exp(0.01Z_i), 0.01, 2), \quad R \leftarrow 0.99R + 0.01R_0 + 0.005E,$$

where Z_i are seeded standard-normal draws, R_0 has off-diagonal entries 0.3, and E is a symmetrized seeded Gaussian perturbation with zero diagonal. The implementation clips correlations to $[-0.99, 0.99]$ and applies a positive-definiteness correction when necessary. In words:

- **Volatilities:** geometric Brownian motion with 1% daily noise
- **Correlations:** a discrete 1% pull toward the 0.3 base-correlation matrix with a 0.5% symmetric daily perturbation, then a positive-definiteness correction if required
- **Weights:** constant (equal-weight portfolio, $w_i = 1/5$)

For each day t , the Eigen-COS algorithm (Nagy, 2026, `fin_exact_var_computational`) computes $A(t) \in \mathbb{R}^{128}$ using $N = 128$ Fourier-cosine terms and an integration domain $[a, b]$ determined by the cumulant-based rule of Fang and Oosterlee (2008). If this domain varies by day, the coefficient index refers to a day-specific basis; accordingly, the reported PCA is a coordinate-level empirical compression conditional on the domain rule, not a fixed-function-space analysis.

2.2 SVD Decomposition

Center the coefficient matrix: $\tilde{A}(t) = A(t) - \bar{A}$. Compute the Singular Value Decomposition:

$$\tilde{A} = U\Sigma V^\top$$

where U is $T \times T$, Σ is diagonal with singular values $\sigma_1 \geq \sigma_2 \geq \dots$, and V is 128×128 . The columns of V are the principal modes v_j , and the coordinates are $z_j(t) = \tilde{A}(t)^\top v_j$.

2.3 Projection and Reconstruction

Projection: given a new coefficient vector A^* , compute

$$z_j = (A^* - \bar{A})^\top v_j, \quad j = 1, \dots, r.$$

Reconstruction: from r coordinates, recover

$$\hat{A} = \bar{A} + \sum_{j=1}^r z_j \cdot v_j.$$

The reconstruction error is $\|A^* - \hat{A}\|_2$.

3. Results

3.1 Variance Explained

Table 1 shows the cumulative variance explained by the first r modes.

Modes (r)	Variance explained	Incremental
1	97.81%	97.81%
2	99.98%	2.17%
3	99.9996%	0.02%
5	100.00%	$< 10^{-4}\%$

A single mode captures 97.8% of observed simulated coefficient variation. Three modes capture 99.9996% — for this coefficient representation, nearly all observed variation. Yield-curve PCA also exhibits a concentrated three-factor structure (Litterman and Scheinkman, 1991), but the two studies concern different feature spaces and do not establish a cross-representation performance ranking.

3.2 Reconstruction Quality

Table 2 shows the VaR reconstruction error for $r = 3$ modes.

Statistic	Coefficient L^2 error	VaR(5%) error
Mean	0.000385	0.005%
Median	0.000314	0.004%

Statistic	Coefficient L^2 error	VaR(5%) error
Max	0.001858	0.0416%

The maximum relative VaR error from using 3 numbers instead of 128 is 0.0416% under the declared simulation protocol. This is a finite, source-bound simulation result; no operational materiality threshold is inferred.

3.3 Mode Interpretation

Mode 1 (z_1 , 97.8% of variance): the dominant mode is strongly associated with simulated VaR(5%) ($r = 0.998$). PCA signs are arbitrary, so this supports an association with the simulated risk profile rather than a signed causal interpretation.

Mode 2 (z_2 , 2.2% of variance): a residual mode orthogonal to the leading coordinate. Its economic interpretation is untested.

Mode 3 (z_3 , 0.02% of variance): an extremely small residual mode. Its economic or tail interpretation has not been quantitatively established.

Remark. The yield curve analogy is instructive. Three PCA modes of the Treasury yield curve are often interpreted as “level, slope, curvature” (Litterman and Scheinkman, 1991). Our first spectral mode has a strong simulated association with risk level, while the interpretations of the remaining modes as asymmetry and tail structure remain hypotheses for future validation.

3.4 Anomaly Detection

The Mahalanobis distance in the 3D z -space is a descriptive outlier score only. Under the current uncalibrated 2.5σ implementation it flags 22 observations in this simulation. This count is not a detection result: no labeled regimes, false-positive control, or out-of-sample detection evaluation is supplied.

3.5 Compression Summary

Quantity	Raw	Compressed	Ratio
Per day	128 coefficients	3 numbers	43×
250 days	32,000 numbers	1,262 numbers	25×
Information loss	—	< 0.05% VaR error	—

3.6 Out-of-Sample Validation

To test whether the learned basis generalizes beyond its training window, we split the 250-day simulation into a training set (days 1–200) and a holdout set (days 201–250). The SVD is computed on the training period only, yielding principal modes $v_1^{\text{train}}, v_2^{\text{train}}, v_3^{\text{train}}$. We then project the holdout days onto this basis and measure reconstruction quality.

Metric	Training days 1–200	Holdout days 201–250
Mean relative VaR(5%) error	0.00191%	0.01095%
Max relative VaR(5%) error	0.01058%	0.01890%

The holdout result is a replayable finite simulation diagnostic, not evidence of real-world generalization or of a practical materiality threshold. The exact protocol, source hashes, and values are recorded in `forge/fin_learned_basis/replay_results_20260809.json`.

3.7 Robustness Scope

The current reproducibility bundle covers only the declared five-asset simulation and its chronological holdout. Earlier exploratory portfolio-size, correlation-regime, crisis, and raw-return comparisons remain preserved in the repository history, but are not source-bound to the current public protocol and are therefore not claims of this version.

Deferred portfolio-size analysis. A future robustness package must expose each scenario, seed, domain rule, and common task metric before it is promoted into the manuscript.

The historical numerical table and accompanying crisis and raw-return comparisons are deliberately withheld here pending that package; their removal is a scope restriction, not deletion of the underlying exploratory artifacts.

4. The Unification Property

The learned basis is not just a compression tool. It offers a **candidate reduced-order coordinate system** for the spectral risk research program, subject to model-specific approximation validation:

4.1 Dynamic spectral update

The temporal compression of spectral coefficients can be studied through PCA. With the learned basis, Dynamic URRT may be approximated by tracking $z_1(t), z_2(t), z_3(t)$ instead of 128 coefficients, subject to validation of the resulting approximation.

4.2 Bayesian live-risk approximation

A candidate reduced-order Bayesian posterior is:

$$P(z_1, z_2, z_3 \mid \text{data}).$$

A 3D Kalman filter is a candidate approximation to a 128D filter. The quoted computational reduction by a factor of $(128/3)^3 \approx 77,000$ applies only if that approximation is validated for the specified model.

4.3 Schrödinger Bridge (Direction 11)

An approximate reduced-order representation of the minimum-entropy path between two distributions is a path in $\mathbb{R}^3$:

$$z(s), \quad s \in [t, t + 1]$$

connecting today’s (z_1, z_2, z_3) to tomorrow’s. Whether this approximation preserves the relevant optimal-transport structure remains to be tested.

4.4 Instanton Paths (Direction 13)

A candidate reduced-order crash scenario is a path in z -space from the current state to a high-VaR state. The corresponding 3D Euler–Lagrange equations require validation against the original 128D problem.

5. Relationship to Yield Curve PCA

The closest analogy in finance is the PCA decomposition of interest rate yield curves (Litterman and Scheinkman, 1991; Dai and Singleton, 2000). Both share the same structure:

Property	Yield curve PCA	Spectral coefficient PCA
Object	Bond yields $y(T)$	Spectral coefficients A_k
Dimension	\$ \$10 maturities	128 modes
Modes needed	3 (level, slope, curve)	3 (empirical simulated coordinates)
Variance by Mode 1	\$ \$85–90%	97.8%
Variance by 3 modes	\$ \$99%	\$ \$100%
Physical interpretation	Interest rate dynamics	Tentative simulated risk-coordinate interpretation

The simulated spectral coefficient representation exhibits a concentrated explained-variance profile. A fair comparison of compressibility with yield curves would require a common task and evaluation criterion.

6. The 130 vs 3 Distinction

A natural question arises: if 3 numbers suffice, why do we need 130?

The answer is that the two representations serve fundamentally different purposes:

Property	130 parameters (URRT)	3 parameters (PCA)
What it encodes	Any distribution, from scratch	Change relative to a known baseline
Prerequisites	None (portfolio data only)	250 days of training + stored basis
Guarantee	Mathematical (Theorem 7, Nagy 2026a)	Empirical (conditional on training regime)
Failure mode	Scope and assumptions of the cited URRT result	Regime change outside training range
Analogy	GPS coordinates (any point on Earth)	“3 blocks left” (requires knowing where you are)

Under the assumptions and scope of the cited URRT result, the 130-parameter representation is claimed to be universal. This paper does not restate or prove that companion result, and its three-mode compression remains empirical and regime-conditional. In a simulated regime change (2008-type crisis), the learned basis becomes stale and the full 130 parameters may be needed until the basis is re-estimated.

The correct interpretation is hierarchical:

1. Under its stated assumptions, the **URRT** result claims that 130 numbers are sufficient for a single snapshot (static).
2. The **PCA** finds that, in the simulated regime studied here, these 130 numbers move predominantly along 3 observed directions (dynamic).
3. In the simulated regime change, the 3-dimensional trajectory leaves the learned subspace, indicating that the basis may need re-estimation; calibrated anomaly detection remains open.

This hierarchy motivates a bridge to Bayesian Live Risk (Direction #7): the 3 PCA coordinates could serve as an approximate state space for a Kalman filter, but the filtering approximation and any diagnostic based on posterior width require separate validation.

7. Verification Status and Scope

This paper reports a numerical simulation study. Its empirical percentages, reconstruction errors, and stress illustrations are not machine-verified mathematical theorems. Earlier development notes associated several elementary PCA identities with legacy Lean artifacts, but the current repository snapshot does not provide a source-bound, independently buildable Lean package for the table previously shown here.

The canonical formal source currently registers the relevant learned-basis names as hypotheses or conditions. That is useful dependency metadata, but it is not evidence that Bessel’s inequality, Eckart–Young optimality, the VaR reconstruction bound, or the Mahalanobis threshold has been derived end-to-end in the current proof system. Accordingly, this version makes no publication-grade formal-verification claim. A future formal claim requires an exact source path and hash, theorem statements that do not assume their conclusions, a successful kernel replay, and an independently checked Lean export.

8. What This Paper Does Not Claim

The study does not establish that real portfolio-loss dynamics are three-dimensional. It does not provide a calibrated anomaly detector, validate Expected Shortfall or other risk functionals, or prove that a projected Bayesian filter or optimal-transport problem preserves the behavior of the original 128-dimensional system. It also does not establish a universal materiality threshold or a publication-grade machine verification of the reported numerical findings.

9. Conclusion

For the simulated coefficient trajectories studied here, the 128 spectral coordinates are effectively three-dimensional over time: PCA finds that a single mode captures 97.8% of observed coordinate variation, and three modes capture 99.9996% (Table 1, Section 3.1). Because the cumulant-based integration domain can vary by day, this is a coordinate-level empirical result conditional on that domain rule, rather than a fixed-function-space conclusion.

The practical implication is a 3-number risk dashboard:

- z_1 = a coordinate strongly associated with simulated VaR
- z_2 = a residual coordinate with tentative economic interpretation
- z_3 = a small residual coordinate with untested tail interpretation

On this simulation, the three coordinates reconstruct $\text{VaR}(5\%)$ within the reported error below 0.05%. Accuracy for ES, other quantiles, moments, and other spectral risk measures remains to be tested.

The learned basis offers a candidate reduced-order representation for the spectral risk program. Dynamic URRT, Bayesian filtering, optimal transport, and instanton analysis require model-specific approximation validation before they can be treated as 3-dimensional problems.

Limitations. The reported evidence is a finite simulation with one five-asset protocol and one chronological holdout (Section 3.6). The daily cumulant-based integration domain can vary, so the compression is a coordinate-level analysis conditional on that domain rule rather than a fixed-function-space result. Validation on real market data, Expected Shortfall and other risk functionals, cross-portfolio robustness, fair comparative benchmarks, and calibrated out-of-sample detection remains essential future work. The 3-mode sufficiency is conditional on the training regime being representative; under a structural break, the basis may need re-estimation. No publication-grade formal verification claim is made for these numerical results.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used Cursor and its integrated AI models (Claude and GPT families) in order to assist with manuscript drafting, structural editing, and language polishing. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

References

- Dai, Q. and Singleton, K. J. (2000). Specification Analysis of Affine Term Structure Models. *The Journal of Finance*, 55(5), 1943-1978. DOI: 10.1111/0022-1082.00278
- Fang, Fang and Oosterlee, Cornelis W. (2008). A Novel Pricing Method for European Options Based on Fourier-Cosine Series Expansions. *SIAM Journal on Scientific Computing*, 31(2), 826-848. DOI: 10.1137/080718061
- Litterman, R. and Scheinkman, J (1991). Common factors affecting bond returns. *Litterman, R. and Scheinkman, J.*, 1(1). DOI: 10.3905/jfi.1991.692347
- Nagy, T. (2026). Deterministic Log-CF Evaluation for Correlated Lognormal Sums. *Zenodo*. DOI: 10.5281/zenodo.21443048
- Nagy, T. (2026). Exact Portfolio VaR Without Monte Carlo: The Eigen-COS Method. *Zenodo*. DOI: 10.5281/zenodo.18910516
- Nagy, T. (2026). Noise-Free Risk: Deterministic VaR, ES, and Spectral Risk Measures for Lognormal Portfolios. *Working paper*.
- Nagy, T. (2026). The Anomaly Functional: Real-Time Arbitrage Detection via Spectral Risk Coefficients. *Working paper*.